Enrollment No.

MID-TERM EXAMINATION
(Course Name : B. Tech) (Semester : 2nd)
(March, 2024) OFFLINE mode

Subject Code: BAI102	**Subject: Object oriented programming using Java**	
Time : 1 ½Hours		**Maximum Marks :30**
Note:Q. 1 is compulsory. Attempt any two parts from Q. 2 and Q. 3.		
Q 1		**(5*2)**
	(a)Explain the role of :i) JVM ii)Constructors	2 + 2½
	(b)Use program explain :i)Multilevel inheritance ii) difference between instance variable,reference variables and objects	2½ + 2
Q 2	**(Attempt any Two Parts)**	**(5,5)**
	(a)Explain use of i) adhoc polymorphism ii) abstract methods	
	(b)Create a Java program that calculates area and perimeter of square and rectangle. program should use function overloading and each function should take as input the required arguments and return both area and perimeter.Also take required input from the users.	5
	(c) Using program explain-: i)Array of objects ii) Instance variable hiding in java	
Q 3	**(Attempt any Two Parts)**	**(5,5)**
	(a)Create a class IgdtuwBooks which describes its Book_title and Book_price. Use getter and setter methods to get & set the Books description. Create object of Book class.Display the books along with its description. Create at least 5 objects of the Book using an array.	
	(b)Explain the use of super and final keywords in java.	
	(c) Compare : i) predefined and user defined package ii) private and protected access specifiers	